Source-style common 2+2 norm in the actual rational-point stalk

Formal proof packet

28 October 2026

This packet records checked algebraic transport and a conditional local-order corollary. It contains no target result, experiment, or cost ratio. The Lean source, compiler log, editable SVG, and CSV gate table accompany this PDF.

Checked boundary

For a rational point $y^2 = f(a)$ on the actual monic quintic chart, the checked ring map sends the quintic $f(X)$ to Y^2 in the weighted-Proj point stalk. Consequently a polynomial identity $n^2 - u_1u_2U = b^2f$ maps to

$$(n(X) - bY)(n(X) + bY) = u_1(X)u_2(X)U(X)$$

in that actual stalk. The source-style common 2+2 theorem supplies this identity under explicit valid Mumford congruences and a nonzero interpolation determinant. This algebraic transport needs neither a DVR nor odd characteristic.

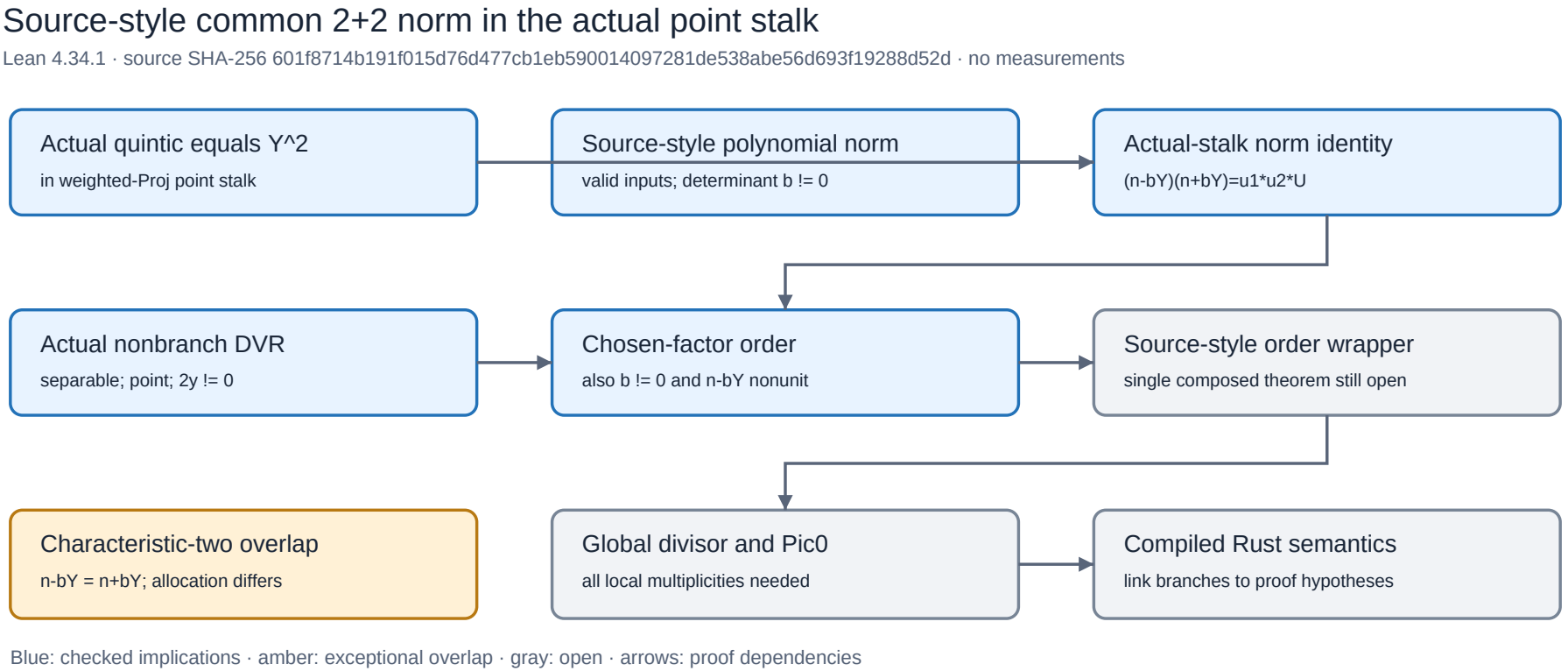
The separate checked `projective_norm_order` theorem uses the actual nonbranch DVR when f is separable, $2y \neq 0$, and $b \neq 0$. If the chosen factor $n(X) - bY$ is a nonunit, the other conjugate factor is a unit, so its order is the sum of the u_1, u_2, U orders. The polynomial identity and chosen-factor nonunit premise are explicit. The source-style identity and generic order theorem have not yet been combined into a single source-style order wrapper.

Exceptional boundary and verification

In characteristic two the conjugate factors coincide, since $n - bY = n + bY$. The norm identity still transports, but this order-allocation argument does not apply even at derivative-smooth points. Branch points with $y = 0$ use the separate checked branch analysis.

Direct pinned Lean 4.34.1 compilation exited zero against Mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612`. Four printed axiom audits list only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Source SHA-256: `601f8714b191f015d76d477cb1eb590014097281de538abe56d693f19288d52d`. The local DVR criterion follows pinned Mathlib `IsDiscreteValuationRing.TFAE` and Stacks Project Tag 00PD. Global principal-divisor/Pic^0 transport and compiled Rust semantics remain open.

Proof dependencies and open gates



Blue nodes are checked implications. Amber marks characteristic-two overlap; gray nodes remain open. Arrows show proof dependencies, not measured performance.